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Mini Project – Final Review

Smart Life

Smart Home with ESP32 Control

Automating Home • Ensuring Safety • Enhancing Life



Project Members :

- Yuvaraj Subramani . M – Reg.No: 24619137
- Syed Mohammed Buhari . S – Reg.No: 24619131
- Mukesh Kumar . B – Reg.No: 24619146

Project Guide :

S. Hariprabha
Assistant Professor

Department of Artificial Intelligence & Machine Learning (AIML)

Email: 24619136@vistas.ac.in * Email: 24619131@vistas.ac.in * Email: 24619146@vistas.ac.in

ABSTRACT

- **Smart Home Automation:** An IoT-based smart home system that enables remote control and monitoring of home appliances like lights, fans and sockets using ESP32 and a web dashboard. The system operates over Wi-Fi for real-time interaction.

- **Built-in Safety System:** Incorporates an MQ-2 gas sensor to detect LPG and smoke. When the gas level exceeds the threshold, the ESP32 triggers a safety relay, blinks an LED indicator, and sends an alert to the dashboard and user.

- **Multi-Channel Control:** Uses a 4-channel relay module to control multiple appliances independently. An IR receiver allows appliance control using an IR remote for added convenience.

- **Local & Secure Operation:** ESP32 hosts a local web server for the dashboard, ensuring no cloud dependency. Access is protected using hashed passwords, CAPTCHA and OTP recovery for secure and reliable operation.

PROBLEM STATEMENT



No Remote Monitoring: Users cannot easily monitor and control home devices from anywhere.



Safety Issues: Gas leakage and other household hazards may not be detected quickly.



High Cost & Complexity: Existing smart-home systems can be expensive and difficult to install.



Need for IoT Automation: A low-cost ESP32-based system is needed for smart control, real-time monitoring, and improved safety.

OBJECTIVE

The objective of this project is to develop a **smart home automation system** using **IoT** and **ESP32**. It allows users to control home appliances like lights and fans through **Wi-Fi**. The system also detects **gas leakage** and gives **safety alerts**. It can control **IR-based appliances** and provides a simple way to monitor the home.





Literature Review

YEAR	AUTHORS	PAPER / METHOD	RELEVANCE TO THIS PROJECT
2023	Vivek Kadiyan et al.	Smart Home Automation System Based on IoT	Supports IoT-based home automation and remote control.
2023	Darko Hercog et al.	Design and Implementation of ESP32-Based IoT Devices	Provides a foundation for ESP32-based IoT systems.
2025	Filippos Serepas et al.	Lightweight Embedded IoT Gateway for Smart Homes Based on an ESP32 Microcontroller	Supports ESP32-based smart-home monitoring and control.

REVIEW TAKEAWAY

The reviewed studies confirm the effectiveness of IoT and ESP32 in enabling smart home automation, remote control, and real-time monitoring.

RESEARCH GAP



- **Limited Remote Control:** Many basic systems do not provide easy control and monitoring of home appliances from anywhere.



- **Safety Limitations:** Gas leakage and other household hazards may not be detected and alerted quickly.



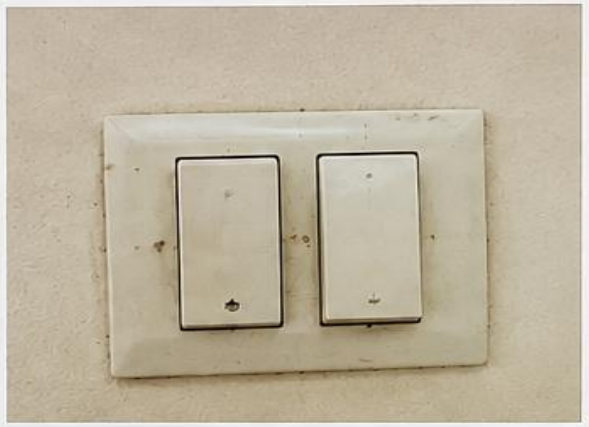
- **High Cost & Complexity:** Existing smart-home solutions can require expensive components and complicated installation.



- **Limited Integration:** There is a need for a simple ESP32-based system that combines appliance control, IR control, gas detection, alerts, and real-time monitoring.

EXISTING SYSTEM

- Home appliances are controlled manually using physical switches and traditional remote controls.
- There is no real-time monitoring of the home environment (e.g., gas leakage).
- Separate remotes are required for different devices such as TV, AC and other appliances.
- There is no centralized system to control multiple appliances together.
- Lack of automation and automatic safety alerts.
- Problems such as gas leakage or devices left ON cannot be detected immediately.
- Not energy efficient and inconvenient for daily use.



Manual Switch Control



Separate Remote for TV



Separate Remote for AC



No Gas Leakage Monitoring

PROPOSED SYSTEM

- ESP32 based smart home automation system to control home appliances (lights, fan, TV/AC) through a web dashboard over Wi-Fi.
- Real-time monitoring of home environment using MQ2 gas sensor.
- Control of multiple appliances using 4-channel relay module.
- IR remote learning and replay to control TV/AC and other IR devices.
- User-friendly web interface with voice command support.
- Automatic safety alerts for gas leakage.
- Low-cost, efficient and integrated system for a smart and safe home environment.



Hardware Prototype (Proposed System)



Web Dashboard (Control & Voice Command)

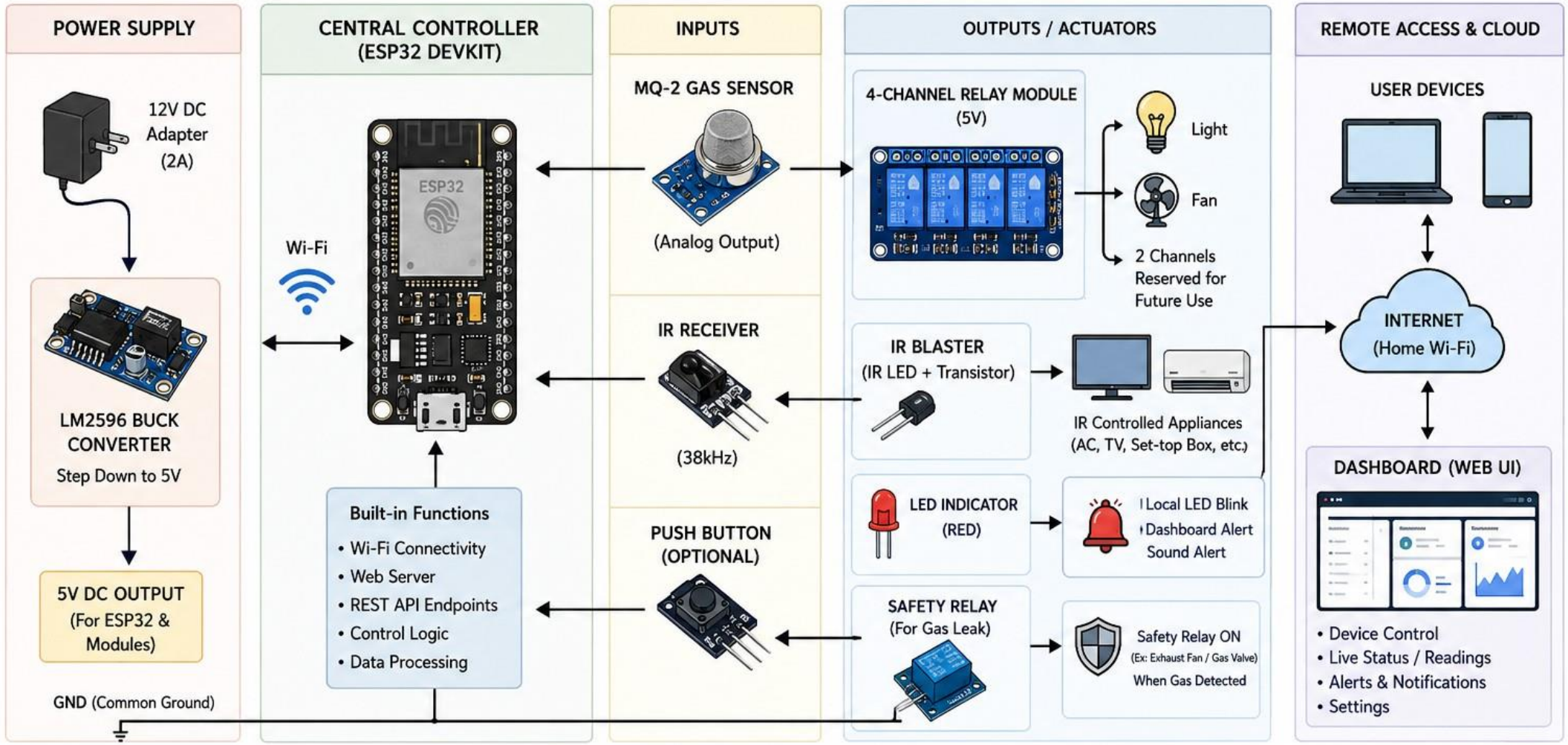


Gas Sensor Monitoring (Safety Alert)



Appliance Control (Light & Fan)

SYSTEM ARCHITECTURE



COMMUNICATION
Wi-Fi (Local Network)
REST API over HTTP

SECURITY
- Hashed Password
- CAPTCHA on Login/Signup
- OTP Based Password Recovery

DATA & STORAGE
- Device & User Settings
- Gas Thresholds
- IR Codes (Learned)
  (Saved in ESP32 Storage)

METHODOLOGY

1 Requirement Analysis

- Identify the home appliances to be controlled (lights, fan, TV/AC).
- Define sensors and control requirements (MQ2 gas sensor, IR, relay).

2 System Design

- Design hardware connections and overall system flow.
- Plan the web dashboard interface and communication.

3 Hardware Implementation

- Interface ESP32 with 4-channel relay module.
- Connect MQ2 gas sensor for gas leak detection.
- Connect IR receiver for learning and replaying remote codes.

4 Software Development

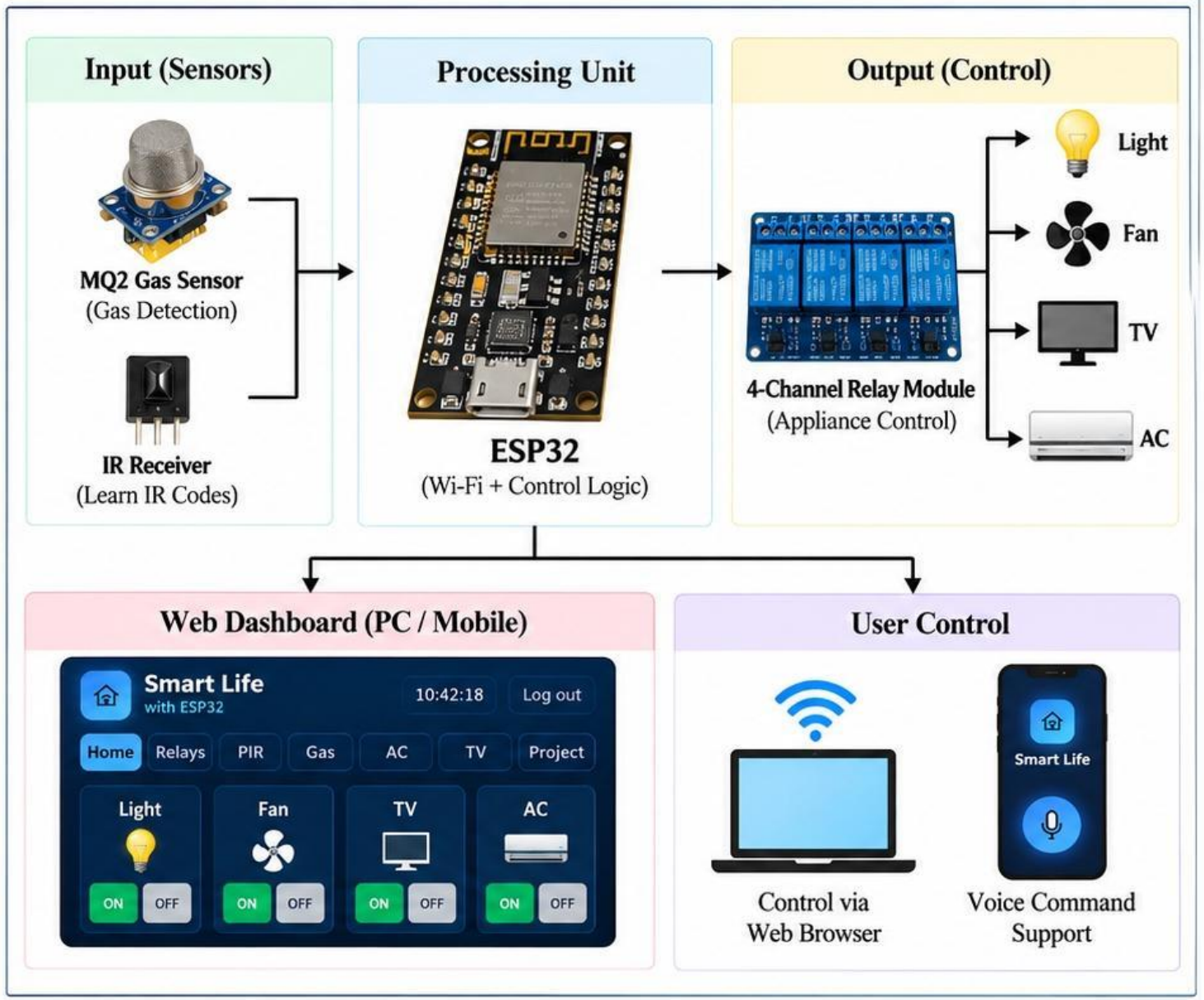
- Program ESP32 (Wi-Fi, web server, relay control, sensor reading, IR functions).
- Develop web dashboard with device control and voice command support.

5 Testing and Validation

- Test individual modules (sensors, relay, IR).
- Integrate and test complete system for real-time control and safety alerts.

6 Deployment and Usage

- Connect ESP32 and user device to the same Wi-Fi network.
- Access the web dashboard to monitor sensors and control appliances.



TECHNOLOGY USED

- **ESP32** – Microcontroller with built-in Wi-Fi used for controlling and monitoring.
- **Arduino IDE** – Used for writing, compiling and uploading the code to ESP32.
- **Arduino C/C++** – Programming language used to develop the logic.
- **Wi-Fi** – Used for wireless communication between ESP32 and web dashboard.
- **HTML** – Used to create the structure of the web dashboard.
- **CSS** – Used for designing the user interface.
- **JavaScript** – Used to add interactivity and real-time control on the web page.
- **ESP32 Web Server** – Hosts the web dashboard locally for device control.
- **MQ-2 Gas Sensor** – Detects LPG, smoke and other gases.
- **IR Receiver** – Receives IR signals for controlling appliances.
- **4-Channel Relay Module** – Controls home appliances (lights, fan, TV, AC).
- **LM2596 Buck Converter** – Steps down the supply voltage to 5V for the relay module.
- **Connecting Wires and Breadboard** – Used for circuit connections and prototyping.

Category	Technology / Tool
Microcontroller	ESP32 Dev Kit (Wi-Fi)
Programming Environment	Arduino IDE
Programming Language	Arduino C / C++
Web Technologies	HTML, CSS, JavaScript
Communication	Wi-Fi (ESP32 Web Server)
Sensors	MQ-2 Gas Sensor, IR Receiver
Actuation	4-Channel Relay Module
Power Supply	LM2596 Buck Converter (5V)
Other Components	Connecting Wires, Breadboard

DATA SET

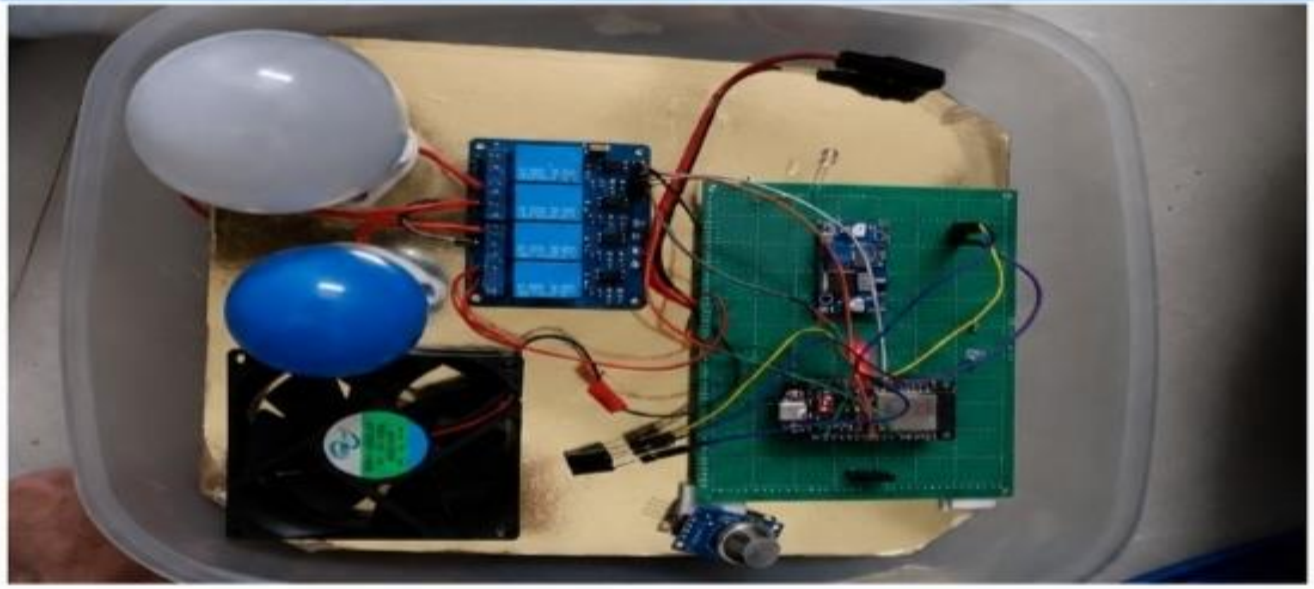
- This project uses real-time data collected from various sensors connected to the ESP32.
- The data includes gas concentration levels, motion detection status and control commands for home appliances.
- The data is generated in real-time during the operation of the smart home system.
- Sensor readings are continuously captured and processed by the ESP32 and displayed on the web dashboard.
- No external dataset is used as this is a real-time IoT project.

Data Type	Description
Gas Sensor Data	MQ-2 sensor readings (analog value) indicating LPG/smoke level in the environment.
Motion Sensor Data	PIR sensor output (High/Low) to detect human movement.
IR Commands	IR receiver data for TV, AC and other device control (decoded IR codes).
Device Status	ON/OFF status of connected appliances (lights, fan, TV, AC).
User Commands	Control commands from the web dashboard (ON/OFF).
Communication Data	Wi-Fi based data exchange between ESP32 and web dashboard.

IMPLEMENTATION (MODULES)

The Smart Life system is implemented using ESP32 as the main controller with various sensors, relay modules, IR module and other components. The hardware components are connected and programmed using Arduino IDE.

Prototype Setup (Assembled)



Components Used



Implementation Details

- All components are mounted on a base board and connected to the ESP32 as per the circuit design.
- Relay module is used to control home appliances (light, fan, etc.).
- Sensors (PIR, LDR, MQ-2) are interfaced with ESP32.
- IR receiver and IR transmitter are connected for device control (TV, AC, projector).
- Power supply is provided to the ESP32 and modules.
- The complete setup is placed in a box for protection and portability.

Main Components Used

- ESP32 Development Board
- 4-Channel Relay Module
- PIR Motion Sensor
- LDR Sensor
- MQ-2 Gas Sensor
- IR Receiver
- IR Transmitter (IR LED)
- Buzzer
- LED
- DC Fan (for demonstration)
- LED Bulb (for demonstration)
- Jumper Wires
- Breadboard / PCB Board
- Power Supply (5V)
- Connecting Wires
- Resistors (as required)
- Mobile / Laptop (for web interface and programming)
- Other basic electronic components

RESULTS



Complete Smart Home Prototype (Actual Result)

Web Dashboard

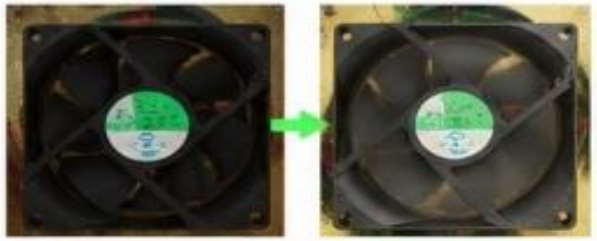


Web Dashboard for Device Control

Device Control Results

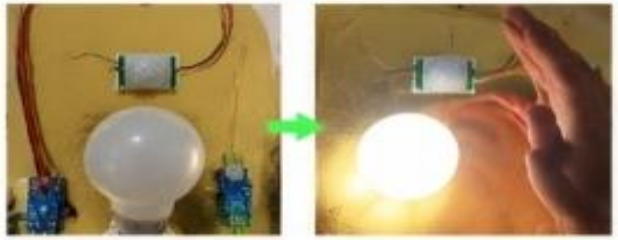


Light OFF | Light ON



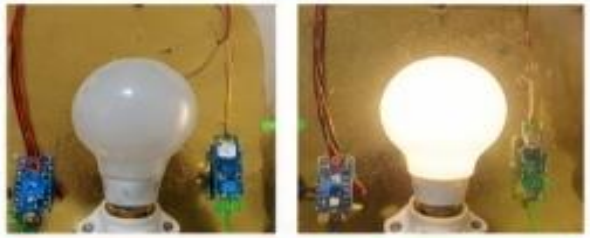
Fan OFF | Fan ON

PIR Sensor Result



No Motion Detected | Motion Detected

LDR Sensor Result



Bright Environment | Dark Environment

MQ-2 Gas Sensor Result



Normal (No Gas) | Gas Detected

IR Control Result



IR Command Sent | TV Controlled

FUTURE ENHANCEMENT

The Smart Life system can be further improved by adding more features and advanced technologies to make it more efficient, secure and user-friendly. Some of the possible future enhancements are listed below:

- **Voice Control Integration:** The system can be integrated with voice assistants such as Google Assistant or Amazon Alexa for hands-free control of home appliances.
- **Mobile Application Development:** A dedicated Android/iOS app can be developed for better user experience and additional features.
- **Real-time Notifications:** Alert messages can be sent to the user's mobile phone or email in case of gas leakage, unauthorized access or any abnormal activity.
- **Advanced Security Features:** Additional sensors such as door/window sensors and camera modules can be included for enhanced home security.
- **AI and Machine Learning:** Smart algorithms can be implemented to learn user behaviour and automate appliances based on daily routines.
- **Energy Monitoring:** Power consumption monitoring can be added to track and reduce energy usage in the home.
- **Solar Power Integration:** The system can be integrated with solar power supply to make it more energy efficient and environmentally friendly.
- **Scalability:** The system can be extended to control more appliances and cover larger areas such as apartments, commercial buildings or industrial spaces.
- **Cloud Integration:** Data can be stored in cloud for remote access, backup and analytics.

CONCLUSION

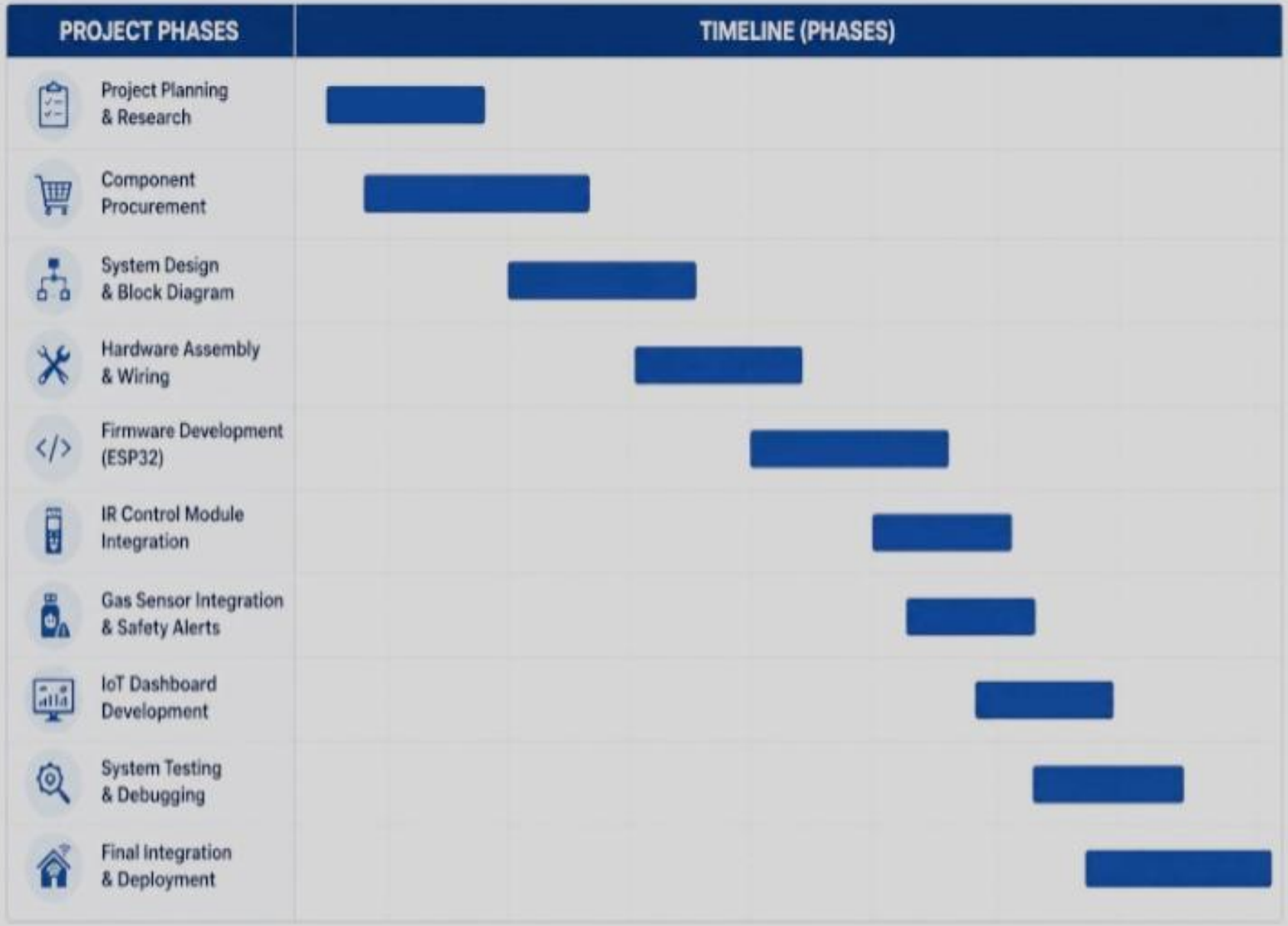
The Smart Life project successfully demonstrates a low-cost and efficient smart home automation system using ESP32. It integrates various sensors and modules to provide automatic and manual control of home appliances through a web dashboard and Wi-Fi connectivity.

- The system can **control lights and fan** automatically based on sensor inputs and manually through a web interface.
- It detects **human motion, ambient light level and gas leakage,** providing real-time monitoring and safety alerts.
- **IR control** enables management of appliances like TV and AC, increasing convenience for the user.
- The project proves that **IoT and ESP32** can be used to build a **smart, secure and energy-efficient** home solution with minimal cost.

Overall, the Smart Life system achieves the main objective of making home automation **simple, affordable and practical** for real-world use, while also providing a strong foundation for future enhancements.

GANTT CHART

IoT Smart Home Automation System - Project Timeline



Thank You